

Module	Unit	Content	Hrs (30+30)	Marks
I		Introduction to Spreadsheet	7	10
	1	Overview of spreadsheet software (e.g., Microsoft Excel, Google Sheets), Basic spreadsheet navigation and interface	1	
	2	Entering and formatting data, Data types and cell formatting, Sorting and filtering data	2	
	3	Arithmetic operations and basic formulas, Common functions (SUM, AVERAGE, MIN, MAX, COUNT)	2	
	4	Text functions (CONCATENATE, LEFT, RIGHT, MID, TEXT), Date and time functions	2	
II		Data Cleaning and Visualization	8	15
	5	Logical functions (IF, AND, OR, NOT, IFERROR, IFS, SWITCH), Lookup and reference functions (VLOOKUP, HLOOKUP, INDEX, MATCH)	2	
	6	Financial functions (NPV, IRR, PMT), Array formulas (SUMPRODUCT, SUMIF, AVERAGE, TRANSPOSE, Array Multiplication, FILTER, IMPORTRANGE)	2	
	7	Handling missing values and duplicates, Data transformation techniques (text to columns, merging cells)	2	
	8	Using advanced text functions for data cleaning, Data validation rules and error-checking	1	
	9	Creating and customizing charts (bar, line, pie, scatter, Histogram), Conditional formatting for data visualization	1	
III		Pivot Tables and Pivot Charts	8	10
	10	Sparklines and data bars, Advanced chart techniques (combination charts, dual-axis charts)	2	
	11	Creating and configuring pivot tables, Grouping and summarizing data in pivot tables	2	
	12	Creating and customizing pivot charts	2	
	13	Using slicers and timeline for interactive analysis	2	
IV		Data Analysis Technique	7	15
	14	Descriptive statistics (mean, median, mode, standard deviation)	1	
	15	Correlation and regression analysis with example, Data analysis tools (Solver, Analysis ToolPak)	2	
	16	Scenario analysis and what-if analysis (Goal Seek, Data Tables, Scenario Manager)	2	
	17	Introduction to DAX (Data Analysis Expressions) for complex calculations (Concept Only)	2	

I. Introduction to Spreadsheet

1. Overview of Spreadsheet Software (e.g., Microsoft Excel, Google Sheets), Basic Spreadsheet Navigation and Interface

1. **Definition:** A spreadsheet is a digital tool used to organize, analyze, and store data in tabular format, consisting of rows and columns.
 2. **Popular Software:** Microsoft Excel (desktop-based) and Google Sheets (cloud-based) are the most widely used spreadsheet applications.
 3. **Basic Navigation:** Includes understanding the workbook, worksheets, rows, columns, cells, and the formula bar.
 4. **User Interface Components:** Ribbon, toolbar, status bar, formula bar, and different sheet tabs for organizing data efficiently.
 5. **Usage in Various Fields:** Used in business, finance, education, and research for data management, analysis, and visualization.
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2. Entering and Formatting Data, Data Types and Cell Formatting, Sorting and Filtering Data

1. **Entering Data:** Data can be entered in individual cells and categorized as text, numbers, or formulas.
 2. **Data Types:** Common types include text, numbers, dates, currency, percentages, and Boolean (TRUE/FALSE) values.
 3. **Cell Formatting:** Formatting tools allow adjusting font, color, alignment, borders, number formats, and conditional formatting for better readability.
 4. **Sorting Data:** Sorting enables arranging data in ascending or descending order based on one or multiple columns.
 5. **Filtering Data:** Filtering helps display only specific data that meets certain criteria, making it easier to analyze large datasets.
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3. Arithmetic Operations and Basic Formulas, Common Functions (SUM, AVERAGE, MIN, MAX, COUNT)

1. **Arithmetic Operations:** Excel allows basic arithmetic operations like addition (+), subtraction (-), multiplication (*), and division (/).
 2. **Basic Formulas:** Formulas begin with = followed by operations or functions, e.g., =A1+B1, =A1*B1.
 3. **SUM Function:** =SUM(A1:A5) adds up all values in the specified range.
 4. **AVERAGE Function:** =AVERAGE(A1:A5) calculates the average (mean) of the selected values.
 5. **MIN, MAX, COUNT Functions:**
 - =MIN(A1:A5) : Returns the smallest value in the range.
 - =MAX(A1:A5) : Returns the largest value in the range.
 - =COUNT(A1:A5) : Counts the number of numerical entries in the range.
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4. Text Functions (CONCATENATE, LEFT, RIGHT, MID, TEXT), Date and Time Functions

1. **CONCATENATE Function** (`=CONCATENATE (A1 , B1 )`): Joins two or more text strings into one. (Note: CONCATENATE is replaced by CONCAT in newer Excel versions.)
2. **LEFT, RIGHT, MID Functions:**
 - `=LEFT (A1, 3)`: Extracts the first three characters from a text string.
 - `=RIGHT (A1, 3)`: Extracts the last three characters.
 - `=MID (A1, 2, 4)`: Extracts four characters starting from the second position.
3. **TEXT Function** (`=TEXT (A1, "dd-mm-yy")`): Converts a value into a specific text format.
4. **Date Functions:**
 - `=TODAY ()`: Returns the current date.
 - `=NOW ()`: Returns the current date and time.
5. **Time Functions:**
 - `=HOUR (NOW ())`: Extracts the hour from the current time.
 - `=MINUTE (NOW ())`: Extracts the minute.
 - `=SECOND (NOW ())`: Extracts the second.

II. Data Cleaning and Visualization

5. Logical Functions (IF, AND, OR, NOT, IFERROR, IFS, SWITCH), Lookup and Reference Functions (VLOOKUP, HLOOKUP, INDEX, MATCH)

Logical Functions:

1. **IF** – `=IF (A1>50, "Pass", "Fail")` returns different results based on a condition.
2. **AND** – Returns TRUE if all conditions are TRUE. Example: `=AND (A1>10, B1<100)`.
3. **OR** – Returns TRUE if any condition is TRUE. Example: `=OR (A1>10, B1<100)`.
4. **NOT** – Reverses the result of a logical test. Example: `=NOT (A1>10)`.
5. **IFERROR** – Returns a specified value if a formula results in an error. Example: `=IFERROR (A1/B1, "Error")`.
6. **IFS** – Tests multiple conditions. Example: `=IFS (A1>90, "A", A1>80, "B")`.
7. **SWITCH** – Checks an expression against a list of values. Example: `=SWITCH (A1, 1, "One", 2, "Two", "Other")`.

Lookup and Reference Functions:

1. **VLOOKUP** – Searches vertically in a table. Example: `=VLOOKUP (101, A2:D10, 2, FALSE)`.
2. **HLOOKUP** – Searches horizontally across rows. Example: `=HLOOKUP ("Math", A1:Z3, 2, FALSE)`.
3. **INDEX** – Returns the value of a cell at a specific row and column. Example: `=INDEX (A2:C6, 2, 3)`.
4. **MATCH** – Returns the position of a value. Example: `=MATCH (45, A1:A10, 0)`.
5. **Used for dynamic data retrieval and combining with other formulas for powerful lookups.**

6. Financial Functions (NPV, IRR, PMT), Array Formulas (SUMPRODUCT, SUMIF, AVERAGE, TRANSPOSE, Array Multiplication, FILTER, IMPORTRANGE)

Financial Functions:

1. **NPV** – Calculates Net Present Value. Example: `=NPV (rate, value1, value2...)`.
2. **IRR** – Returns the Internal Rate of Return. Example: `=IRR (values)`.
3. **PMT** – Calculates loan payments. Example: `=PMT (rate, nper, pv)`.

Array Formulas:

1. **SUMPRODUCT** – Multiplies corresponding array elements and returns the sum. `=SUMPRODUCT (A1 : A3, B1 : B3)`.
 2. **SUMIF** – Adds values that meet a condition. Example: `=SUMIF (A1 : A5, ">50")`.
 3. **AVERAGE** – Calculates the mean of selected cells. `=AVERAGE (A1 : A10)`.
 4. **TRANSPOSE** – Changes rows to columns and vice versa. Example: `=TRANSPOSE (A1 : B3)`.
 5. **Array Multiplication** – Used to multiply values across ranges.
 6. **FILTER** – Filters a range based on criteria. Example: `=FILTER (A2 : B10, B2 : B10 > 50)`.
 7. **IMPORTRANGE** – Imports data from another Google Sheet. Example: `=IMPORTRANGE ("URL", "Sheet1!A1 : B10")`.
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7. Handling Missing Values and Duplicates, Data Transformation Techniques (Text to Columns, Merging Cells)

1. **Handling Missing Values** – Use functions like `IF (ISBLANK (A1), "Missing", A1)` to identify or fill gaps.
 2. **Removing Duplicates** – Excel provides a tool under the Data tab to remove duplicate entries from selected columns.
 3. **Text to Columns** – Splits text from one cell into multiple columns using delimiters like comma or space.
 4. **Merging Cells** – Combines two or more cells into one for formatting purposes (not suitable for data analysis).
 5. **Cleaning Data** – Includes trimming spaces, correcting capitalization, and standardizing values for consistency.
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8. Using Advanced Text Functions for Data Cleaning, Data Validation Rules and Error-Checking

1. **TRIM** – Removes extra spaces. Example: `=TRIM (A1)`.
 2. **PROPER, UPPER, LOWER** – Adjusts text case. `=PROPER (A1)` capitalizes first letters.
 3. **SUBSTITUTE and REPLACE** – Replaces or modifies text. `=SUBSTITUTE (A1, "old", "new")`.
 4. **Data Validation** – Restricts input (e.g., only numbers, specific ranges) to maintain data integrity.
 5. **Error Checking** – Tools like `IFERROR` or `ISERROR` help catch and manage errors in formulas.
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9. Creating and Customizing Charts (Bar, Line, Pie, Scatter, Histogram), Conditional Formatting for Data Visualization

1. **Bar and Column Charts** – Compare different categories using horizontal or vertical bars.
2. **Line Chart** – Shows trends over time. Useful for time-series data.
3. **Pie Chart** – Displays parts of a whole as slices.
4. **Scatter Plot** – Shows relationships between two variables (e.g., X and Y values).
5. **Histogram** – Displays frequency distribution of numerical data.
6. **Conditional Formatting** – Visually highlights cells based on rules (e.g., red for values < 50).
7. **Customization** – Titles, legends, labels, and color schemes can be adjusted for better readability.

III. Pivot Tables and Pivot Charts

10. Sparklines and Data Bars, Advanced Chart Techniques (Combination Charts, Dual-Axis Charts)

1. **Sparklines** – Tiny charts inside a single cell to show trends. Types include Line, Column, and Win/Loss.
2. **Usage** – Useful for tracking changes over time, such as sales growth or stock trends.
3. **Data Bars** – A visual representation of data inside a cell where longer bars indicate higher values.
4. **Adding Sparklines** – Found under the *Insert* tab, selecting *Sparklines* and choosing data range.
5. **Customizing Sparklines** – Change color, highlight high/low points, and adjust line style.

Advanced Chart Techniques

1. **Combination Charts** – Mix two or more chart types, e.g., a bar and a line chart together.
 2. **Dual-Axis Charts** – Two vertical axes used when data has different units or ranges.
 3. **Usage** – Helps compare multiple datasets effectively, such as revenue vs. growth percentage.
 4. **Creating in Excel** – Select data, go to *Insert > Chart*, choose *Custom Combination Chart*.
 5. **Customization** – Adjust axis labels, colors, legends, and chart styles for better clarity.
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11. Creating and Configuring Pivot Tables, Grouping and Summarizing Data in Pivot Tables

Creating and Configuring Pivot Tables

1. **Definition** – A pivot table is an interactive table used to summarize, analyze, and explore large datasets.
2. **Steps to Create** – Select data, go to *Insert* tab, click *PivotTable*, choose range and location.
3. **Fields in Pivot Table** – Drag data fields into Rows, Columns, Values, and Filters sections.
4. **Customization** – Change calculation type (Sum, Count, Average), apply filters, and sort data.
5. **Refreshing Data** – If the source data changes, click *Refresh* to update the pivot table.

Grouping and Summarizing Data in Pivot Tables

1. **Grouping Data** – Allows grouping by date, month, quarter, or numeric ranges for better analysis.
 2. **Steps** – Right-click on a field (e.g., date column) and select *Group* to organize data.
 3. **Summarizing Values** – Change calculation methods such as Sum, Average, Min, Max, Count.
 4. **Sorting Data** – Click on column headers to sort in ascending or descending order.
 5. **Filtering Data** – Apply filters or slicers to display specific information dynamically.
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12. Creating and Customizing Pivot Charts

1. **Definition** – Pivot Charts are graphical representations of Pivot Tables, making data analysis easier.
 2. **Steps to Create** – Select Pivot Table, go to *Insert*, choose *Pivot Chart* and select a chart type.
 3. **Types of Pivot Charts** – Column, Bar, Line, Pie, Scatter, etc.
 4. **Customizing Charts** – Modify axis titles, data labels, chart styles, and colors for better visualization.
 5. **Dynamic Updates** – Pivot Charts update automatically when the Pivot Table data changes.
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13. Using Slicers and Timeline for Interactive Analysis

Using Slicers

1. **Definition** – A slicer is a visual filter that allows users to filter data dynamically.
2. **Steps to Add** – Select Pivot Table, go to *Insert > Slicer*, choose the field to filter.
3. **Multiple Selections** – Click multiple slicer buttons to filter data across different categories.
4. **Formatting Slicers** – Customize colors, sizes, and button arrangements for a better look.
5. **Advantages** – Provides an easy and intuitive way to filter Pivot Table or Pivot Chart data.

Using Timeline

1. **Definition** – A timeline is a special slicer used for filtering Pivot Tables based on date ranges.
2. **Steps to Add** – Click Pivot Table, go to *Insert > Timeline*, select a date-based field.
3. **Filtering Data by Period** – Drag the slider to filter data by months, quarters, or years.
4. **Customization** – Change style, layout, and color of the timeline for better readability.
5. **Usage** – Helps track trends over time in financial reports, sales data, or project timelines.

IV. Data Analysis Technique

14. Descriptive Statistics (Mean, Median, Mode, Standard Deviation)

Mean (Average)

1. **Definition** – The mean is the average value of a dataset, calculated by dividing the sum of all values by the number of observations.
2. **Formula** – Mean = (Sum of all values) ÷ (Total number of values).
3. **Usage** – Helps in understanding the central tendency of data in sales, finance, and other fields.
4. **Excel Function** – =AVERAGE (range) .
5. **Limitation** – Affected by extreme values (outliers), which can distort the result.

Median

1. **Definition** – The median is the middle value when data is arranged in ascending or descending order.
2. **Calculation** – If there is an odd number of values, the median is the middle number; if even, it is the average of the two middle numbers.
3. **Usage** – Used in income distribution and test scores to find central value without being affected by outliers.
4. **Excel Function** – =MEDIAN (range) .
5. **Advantage** – Provides a better measure of central tendency when data has extreme values.

Mode

1. **Definition** – The mode is the most frequently occurring value in a dataset.
2. **Types** – A dataset can have one mode (unimodal), two modes (bimodal), or multiple modes (multimodal).
3. **Usage** – Useful in retail for identifying the most popular product or customer preference.
4. **Excel Function** – =MODE (range) .
5. **Limitation** – If no value repeats, there is no mode; if multiple values repeat, it can be difficult to interpret.

Standard Deviation

1. **Definition** – Measures the spread or dispersion of a dataset relative to its mean.
2. **Formula** – The square root of the variance, calculated as:
$$\sigma = \sqrt{(\sum (x_i - \mu)^2 / N)}$$
, where μ is the mean, x_i are data points, and N is the number of observations.
3. **Usage** – Helps assess risk in finance, stock market fluctuations, and quality control in manufacturing.
4. **Excel Function** – =STDEV (range) .

5. **Interpretation** – A low standard deviation indicates that data points are close to the mean, while a high standard deviation suggests a large spread.
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15. Correlation and Regression Analysis with Example, Data Analysis Tools (Solver, Analysis ToolPak)

Correlation Analysis

1. **Definition** – Correlation measures the strength and direction of a relationship between two variables.
2. **Types** – Positive correlation (both variables increase), negative correlation (one increases while the other decreases), and no correlation.
3. **Excel Function** – `=CORREL(range1, range2)`.
4. **Example** – The relationship between advertising spend and sales revenue.
5. **Limitations** – Correlation does not imply causation; a third factor may be influencing both variables.

Regression Analysis

1. **Definition** – Regression estimates the relationship between a dependent variable and one or more independent variables.
2. **Types** – Simple Linear Regression (one independent variable) and Multiple Regression (two or more independent variables).
3. **Formula** – $Y = a + bX$ (where Y is the dependent variable, X is the independent variable, a is the intercept, and b is the slope).
4. **Example** – Predicting future sales based on past advertising expenses.
5. **Excel Tool** – Performed using the *Regression* tool in the *Analysis ToolPak*.

Data Analysis Tools

Solver

1. **Definition** – Solver is an Excel add-in used for optimization problems, such as maximizing profits or minimizing costs.
2. **Usage** – Helps in supply chain management, resource allocation, and financial modeling.
3. **Steps to Use** – Enable Solver from *Excel Options > Add-ins*, define objective, set constraints, and run Solver.
4. **Example** – Finding the best combination of products to maximize profit with a limited budget.
5. **Limitation** – Works best with well-defined problems; complex models may require advanced programming.

Analysis ToolPak

1. **Definition** – An Excel add-in that provides advanced statistical and engineering analysis tools.
 2. **Features** – Includes tools for descriptive statistics, correlation, regression, and hypothesis testing.
 3. **Steps to Enable** – Go to *File > Options > Add-ins > Manage Excel Add-ins > Enable Analysis ToolPak*.
 4. **Usage** – Used in academic research, finance, and quality control.
 5. **Advantage** – Saves time by automating complex calculations.
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16. Scenario Analysis and What-If Analysis (Goal Seek, Data Tables, Scenario Manager)

Scenario Analysis

1. **Definition** – A technique used to analyze different possible future outcomes by changing input values.
2. **Usage** – Helps businesses plan for best-case, worst-case, and average scenarios.

3. **Excel Feature** – Managed using the *Scenario Manager* tool under the *Data* tab.
4. **Example** – Estimating revenue for different market conditions.
5. **Advantage** – Helps in risk management and decision-making.

Goal Seek

1. **Definition** – Goal Seek is used to find the required input value that results in a specific output.
2. **Steps to Use** – Go to *Data > What-If Analysis > Goal Seek*, set the target cell, and define the desired value.
3. **Example** – Finding the required sales price to reach a profit target.
4. **Limitations** – Works with a single input-output relationship only.
5. **Advantage** – Quick and easy to use for simple forecasting problems.

Data Tables

1. **Definition** – Data Tables allow testing different input values to see how they affect results.
2. **Types** – One-variable (changing a single input) and two-variable data tables (changing two inputs).
3. **Example** – Calculating loan payments for different interest rates and loan amounts.
4. **Steps to Use** – Select data, go to *Data > What-If Analysis > Data Table*, define input range.
5. **Advantage** – Helps in financial modeling and scenario planning.

Scenario Manager

1. **Definition** – A tool that allows saving multiple sets of input values and switching between them.
2. **Usage** – Useful for budgeting, investment analysis, and sales forecasting.
3. **Steps to Use** – Go to *Data > What-If Analysis > Scenario Manager*, create and save scenarios.
4. **Example** – Analyzing company profits under different economic conditions.
5. **Advantage** – Allows quick comparisons of multiple business strategies.

17. Introduction to DAX (Data Analysis Expressions) for Complex Calculations (Concept Only)

1. **Definition** – DAX is a formula language used in Power BI, Excel Power Pivot, and Analysis Services for data modeling.
2. **Usage** – Helps in performing advanced calculations on large datasets.
3. **Functions in DAX** – Includes SUMX, AVERAGEX, CALCULATE, and FILTER for complex analysis.
4. **Example** – Calculating running totals or comparing sales growth over different periods.
5. **Advantage** – Enhances data manipulation and reporting capabilities in Excel and Power BI.

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